

PlantHealthMonitor

Manual

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Introduction

PlantHealthMonitor (PHM) is an intelligent plant monitoring and care system that combines embedded IoT hardware with a cross-platform mobile application. The system collects environmental sensor data from a plant's surroundings, classifies plant health in real time using on-device machine learning, and pushes actionable care recommendations to your phone or desktop.

This manual covers everything you need to set up the hardware, install the app, configure your plants, and understand the information displayed on screen.

Key Features

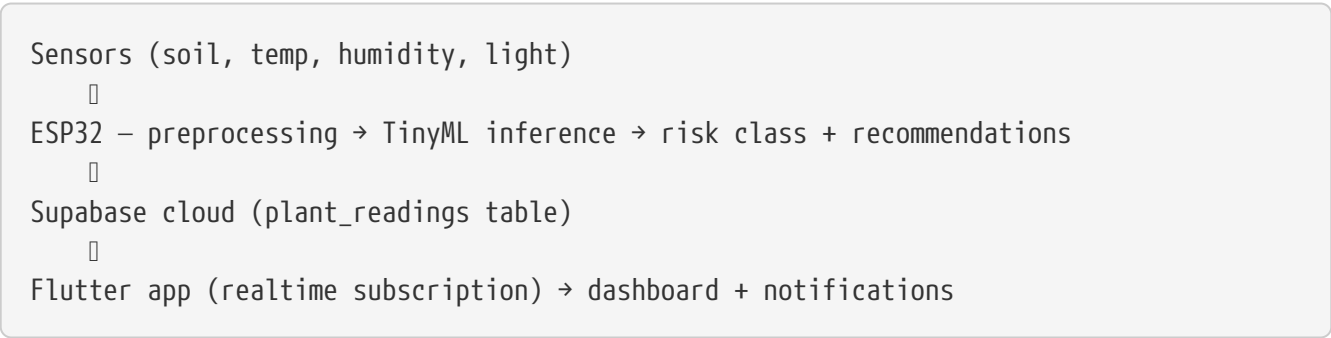
- **Real-time sensor monitoring** — soil moisture, temperature, humidity, and light intensity.
- **On-device ML classification** — a trained neural network (MLP) running on the ESP32 classifies plant health as healthy, moderate, or high risk without a cloud round-trip.
- **ML-powered time predictions** — a second regression model predicts how many minutes until a care action (e.g. watering) becomes urgent.
- **Rule-based care actions** — the system applies sensor thresholds to decide which actions to recommend (water, reduce temperature, increase light, increase humidity).
- **Push notifications** — instant alerts when a plant needs attention, plus an optional daily summary.
- **Multi-plant support** — manage several plants, each with its own device and preference profile.
- **Cross-platform app** — iOS, and Windows are supported.

System Overview

PlantHealthMonitor consists of three layers that work together:

Layer	Component	Responsibility
Edge	ESP32 microcontroller + sensors	Collects sensor readings, runs TinyML inference, syncs data to the cloud every 3 hours.
Cloud	Supabase (PostgreSQL + Realtime)	Stores sensor readings and plant settings; delivers live updates to connected apps.
App	Flutter (Android / iOS / Windows)	Displays live readings, charts, and recommendations; sends push notifications; stores your preferences.

Data Flow



Health Risk Levels

The system classifies each reading into one of three risk levels:

Level	Label	Meaning
0	Healthy	All sensor values fall within your plant’s preferred bands. No action needed.
1	Moderate Risk	One or more values are outside the preferred band but not critical. Monitor closely.
2	High Risk	Critical conditions detected. Immediate action is recommended.

Hardware Setup

Bill of Materials

Qty	Item	Notes
1	ESP32 development board	Any standard 38-pin variant
1	Capacitive soil moisture sensor	Analog output, connects to GPIO 34
1	DHT22 temperature & humidity sensor	Connects to GPIO 4
1	LDR (light-dependent resistor)	Analog output with 10 k Ω pull-down, connects to GPIO 35
1	10 k Ω resistor	For the LDR voltage divider
—	Jumper wires	Male-to-male and male-to-female
1	USB-A to micro-USB or USB-C cable	For flashing and serial monitor

Wiring Diagram

Sensor pin	ESP32 pin
Soil moisture — AOUT	GPIO 34
DHT22 — DATA	GPIO 4
DHT22 — VCC	3.3 V
DHT22 — GND	GND
LDR — one leg (with 10 k Ω to GND)	GPIO 35
LDR — other leg	3.3 V

NOTE

GPIO 34 and 35 are input-only pins on the ESP32. Do not connect any output signal to them.

Flashing the Firmware

The firmware is managed with PlatformIO.

1. Install **PlatformIO** (VS Code extension or CLI).
2. Navigate to the firmware source:

```
cd src/prototype-non-local
```

3. Copy the example configuration and fill in your credentials:

```
cp platformio.ini.example platformio.ini
```

4. Open `platformio.ini` and set the following values:

```
build_flags =
-D WIFI_SSID=\"YourNetworkName\"
-D WIFI_PASSWORD=\"YourPassword\"
-D PLANT_LABEL=\"my_plant\"
-D DEVICE_NAME=\"Living Room\"
```

5. Build and upload to the device (replace `esp32_com3_non_local` with the environment matching your COM port):

```
platformio run -e esp32_com3_non_local
platformio run -t upload -e esp32_com3_non_local
```

6. Verify the device is running correctly:

```
platformio device monitor -b 115200
```

You should see sensor readings and WiFi connection logs printed every 3 hours.

TIP

If you have multiple plants, repeat these steps for each ESP32 with a unique `PLANT_LABEL` value. Available environments are `esp32_com3_non_local`, `esp32_com4_non_local`, and `esp32_com5_non_local`.

Mobile App Setup

Installation

iOS

Sideloady is a free Windows tool that installs IPA files onto an iOS device without a paid Apple Developer account.

Requirements:

- Windows 10 or later.
- iTunes installed (the full installer from Apple's website, not the Microsoft Store version).
- iCloud for Windows installed.
- Your iPhone or iPad connected via USB with the screen unlocked and the device trusted on the PC.

Steps:

1. Download and install [Sideloady](#) on your Windows machine.
2. Download `phm_app_<version>.ipa` from the `releases/` folder.
3. Open Sideloady.
4. In the **Apple Account** field, enter your Apple ID and password.

NOTE

Sideloady uses your Apple ID only to sign the app locally. It is recommended to use a secondary Apple ID rather than your primary account.

5. Drag and drop `phm_app_<version>.ipa` onto the Sideloady window, or click the IPA field and browse to the file.
6. Make sure your device appears in the **Device** dropdown. If it does not, reconnect the USB cable and unlock the device.
7. Click **Start**.
8. If prompted on your iPhone/iPad with "Trust This Computer", tap **Trust**.
9. Wait for Sideloady to finish. A success message will appear in the log panel.
10. On your iOS device, go to **Settings** → **General** → **VPN & Device Management**, find your Apple ID under **Developer App**, and tap **Trust**.
11. Open PlantHealthMonitor from your home screen.

IMPORTANT

Free Apple ID accounts expire the app signature after 7 days. You will need to re-sideload the app to continue using it. A paid Apple Developer account (\$99/year) extends this to 1 year.

Windows

1. Download the release archive from [releases/phm_app_windows_<version>.zip](#).
2. Extract the archive and run [phm_app.exe](#).

Building from Source

```
cd src/phm_app
flutter pub get
flutter run                # connected device or emulator
flutter run -d windows     # Windows desktop
flutter build apk          # Android release APK
```

Flutter 3.x or later is required.

Using the App

Dashboard

The dashboard is the main screen. It shows a card for each plant found in the cloud, including:

- **Status badge** — Healthy / Moderate Risk / High Risk.
- **Last updated** — how long ago the most recent reading arrived.
- **Sensor tiles** — four tiles showing the current values for soil moisture, temperature, humidity, and light.
- **Recommendation summary** — a short text describing any care action needed and the estimated time until it is required.

Tap a sensor tile to see which preference band the current reading falls into (low / mid / high).

Pull down on the dashboard to force a refresh from the cloud.

Plant Selector

If you have more than one plant, use the dropdown at the top of the dashboard to switch between them. Each plant is identified by its **PLANT_LABEL** from the firmware.

Sensor Detail

Tapping the **Data** button on a plant card opens the sensor history charts. Charts show the last week of readings (up to 56 data points at 3-hour intervals) for each sensor.

Plant Preferences

Tap the **Preferences** button (or navigate via the menu) to edit the care preferences for a plant.

Each of the four sensors has a preference selector with three options:

Option	Meaning
Low	The plant prefers lower values for this sensor (e.g. a cactus for soil moisture).
Mid	The plant prefers moderate values.
High	The plant prefers higher values (e.g. a tropical plant for humidity).

After editing, tap **Save**. The preferences are synced to the cloud and used by the recommendation engine on the next sensor reading.

Notifications

Instant Alerts

The app sends a push notification any time a plant's risk class changes to Moderate or High Risk. Duplicate notifications are suppressed — you will not receive the same alert twice for the same condition.

Daily Summary

You can enable a daily summary notification that reports the status of all your plants at a chosen time. To configure this:

1. Open **Settings** from the navigation menu.
2. Toggle **Daily Report** on.
3. Pick a time using the time picker.

The app schedules this notification locally on your device and it fires even if the app is in the background.

Disabling Notifications

Toggle **Instant Alerts** or **Daily Report** off individually in **Settings**.

Understanding Readings & Recommendations

Sensor Ranges

The sensors report percentages (0–100 %) except temperature, which is in degrees Celsius.

Sensor	Low band	Mid band	High band
Soil moisture (%)	0–33	34–66	67–100
Temperature (°C)	< 18	18–28	> 28
Humidity (%)	0–40	41–70	71–100
Light (%)	0–33	34–66	67–100

Recommendation Actions

The recommendation engine can trigger four care actions:

Action	What it means
Water the plant	Soil moisture is below the preferred band or declining towards it. The time estimate shows how long until watering becomes urgent.
Reduce temperature	Temperature is above the preferred band.
Increase light	Light level is below the preferred band.
Increase humidity	Humidity is below the preferred band.

The timing estimate (e.g. "water in 4 hours") is produced by a trained regression neural network running on the ESP32 — it predicts how many minutes until the action becomes urgent based on sensor history. The action flags themselves (water / reduce temp / etc.) are determined by rule-based threshold checks on the raw sensor values, combined with the ML risk class.

When the risk class is 0 (Healthy), no actions are triggered and the recommendation summary shows a positive status message.

Troubleshooting

ESP32 is not connecting to WiFi

- Check that `WIFI_SSID` and `WIFI_PASSWORD` are correct in `platformio.ini` and rebuild.
- The ESP32 only supports 2.4 GHz networks. Make sure you are not connecting to a 5 GHz band.
- Open the serial monitor (`platformio device monitor -b 115200`) to read detailed connection logs.

No data appears in the app

- Confirm the ESP32 is powered on and has successfully connected to WiFi (check the serial monitor).
- The device syncs every 3 hours. Wait for the next sync cycle or reset the device to force an immediate sync.
- Check that the `PLANT_LABEL` in the firmware matches what the app is looking for.

App shows stale data

- Pull down on the dashboard to force a cloud refresh.
- Check your internet connection on the mobile device.

Notification not received

- Make sure notifications are enabled for the app in your device's system settings.
- On Android 13 and later, the app requires the `POST_NOTIFICATIONS` runtime permission. Grant it via **Settings** → **Apps** → **PlantHealthMonitor** → **Permissions**.

Preferences are not saving

- The app requires an internet connection to sync preferences to the cloud. Connect to the internet and try again.
- If the problem persists, force-quit the app and re-enter your preferences.

Technical Reference

Cloud Database Schema

plant_readings

Column	Type	Description
plant_label	text	Unique identifier for the plant (matches firmware PLANT_LABEL).
device_id	text	Hardware identifier of the ESP32.
device_name	text	Human-readable device name.
soil_moisture_pct	float	Soil moisture as a percentage (0–100).
temperature_c	float	Ambient temperature in °C.
humidity_pct	float	Relative humidity as a percentage (0–100).
light_level_pct	float	Light intensity as a percentage (0–100).
risk_class	int	Health classification: 0 = healthy, 1 = moderate, 2 = high.
action_water	bool	Whether watering is recommended.
action_reduce_temp	bool	Whether reducing temperature is recommended.
action_increase_light	bool	Whether increasing light is recommended.
action_increase_humidity	bool	Whether increasing humidity is recommended.
recommendation_summary	text	Human-readable recommendation including time predictions.
timestamp	timestampz	Reading time in UTC.

plant_settings

Column	Type	Description
plant_label	text	Unique plant identifier (primary key).
soil_preference	text	low , mid , or high .
temperature_preference	text	low , mid , or high .
humidity_preference	text	low , mid , or high .
light_preference	text	low , mid , or high .
version	int	Incremented on each save; used for synchronisation.

Firmware Configuration Reference

Flag	Default	Description
WIFI_SSID	—	2.4 GHz network name.
WIFI_PASSWORD	—	Network password.
PLANT_LABEL	—	Unique plant identifier string.
DEVICE_NAME	—	Friendly name shown in the app.

ML Models

The firmware contains two trained neural networks that run directly on the ESP32:

Health risk classifier — a multi-layer perceptron (MLP) with a hidden layer (ReLU activation) and a softmax output layer. It takes normalised sensor features as input and outputs a risk class (0, 1, or 2) with per-class confidence scores. A rule-based fallback classifier is also available and can be selected via the pluggable **MLayer** backend. Edge Impulse models are additionally supported.

Predictive irrigation model — a regression MLP with a hidden layer (ReLU) and a linear output. It was trained on log-transformed target values (minutes until watering is needed) and uses **expm1** to convert the output back to minutes. This value drives the time estimate in the recommendation summary.

The care action flags (water / reduce temp / increase light / increase humidity) are generated by a separate rule-based **RecommendationEngine** that compares raw sensor readings against the per-plant preference thresholds, using the ML risk class as an additional input.

Future Possibilities

This section outlines improvements and extensions that could be added to PlantHealthMonitor in future iterations.

Wireless Configuration WiFi/BLE

Currently, WiFi credentials and the plant label must be compiled into the firmware before flashing. A more user-friendly approach would be to use a WiFi provisioning flow:

- On first boot, the ESP32 broadcasts its own temporary access point.
- The user connects to that access point with their phone and opens a simple configuration page.
- WiFi credentials, plant label, and device name are entered and saved to non-volatile storage on the device.
- The ESP32 then restarts and connects to the configured network.

This would remove the need for PlatformIO entirely for end users and make it practical to move a device to a different network without reflashing. Frameworks such as ESP-IDF's `wifi_provisioning` component or the Arduino `WiFiManager` library already provide most of this functionality.

As an alternative to the WiFi provisioning flow above, Bluetooth Low Energy (BLE) could be used to configure the device from the Flutter app directly. The app would scan for nearby unconfigured devices, connect over BLE, and push the network credentials and plant label — no access point switching required.

BLE could also serve as a local data channel for users who do not want to connect the device to the internet at all, syncing readings directly to the phone when in range.

Built-in Plant Profiles

At the moment, users set preferences manually (low / mid / high per sensor). A future version could ship a plant species library with pre-defined care profiles. Users would pick their plant from a list (e.g. "Snake Plant", "Peace Lily", "Basil") and the correct preferences would be populated automatically. The library could be:

- Bundled with the app for offline use.
- Pulled from a community-maintained cloud database so new species can be added without an app update.

User-defined custom profiles would remain available for unlisted species.

Support for Additional Sensor Types

The current sensor set covers the most common care factors, but the hardware layer could be extended to support:

- **pH sensor** — soil acidity monitoring, relevant for sensitive species.
- **EC (electrical conductivity) sensor** — measures nutrient levels in the soil.
- **CO₂ sensor** — useful for indoor grow environments.

The pluggable **MLayer** architecture already makes it relatively straightforward to retrain models with additional input features once new sensors are added.

Improved ML Model Training Pipeline

The current ML models are trained offline and their weights are exported as C++ header files. A more scalable approach would be to:

- Continuously collect labelled sensor data from all deployed devices (with user consent).
- Retrain the models periodically in the cloud.
- Push updated weights to devices over OTA.

This would allow the models to improve over time and adapt to a wider variety of plants and environments.

Multi-User and Sharing Support

Currently each installation is tied to a single Supabase project. Adding an authentication layer would allow:

- Multiple users to share ownership of the same plant (e.g. housemates).
- Read-only guest access so a plant-sitter can monitor status without changing settings.
- Per-user notification preferences independent of device configuration.

Appendix : Glossary

Term	Definition
ESP32	Low-cost Wi-Fi + Bluetooth microcontroller used as the sensing device.
TinyML / TFLM	TensorFlow Lite Micro — a framework for running ML models on microcontrollers.
Plant label	A string identifier that links a hardware device to its cloud records and app preferences.
Risk class	An integer (0, 1, or 2) output from the ML model indicating the current health level.
Preference band	The user-selected range (low / mid / high) for a sensor that the recommendation engine uses to evaluate readings.
Supabase	The managed cloud backend providing PostgreSQL storage and realtime change events.
OTA	Over-the-Air firmware update — allows flashing new firmware over WiFi without a USB cable.